AI Past Paper - Filtered Syllabus Questions

This document contains only the questions from the provided past papers that relate to your current syllabus (Decision Trees, ID3, Entropy, Information Gain, etc.). Unrelated topics like Genetic Algorithms and Predicate Calculus have been removed.

Part I: Objective Questions (MCQs & True/False)

14. Information gain metric in decision tree induction measures:

(A) Gain in information of the dataset

(B) Reduction in entropy due to partitioning by an attribute **(Correct Answer)**

(C) Entropy of an attribute

(D) Gain in entropy of the dataset

Topic: Lecture 2 - Information Gain

15. An attribute in a categorical dataset that has many values, e.g., Date, has:

(A) Low entropy and high gain

(B) High entropy and low gain

(C) Low entropy and low gain

(D) High entropy and high gain **(Correct Answer)**

Topic: Lecture 2 & 3 - ID3 Splitting and Bias

16. Maximum entropy of a dataset having three classes is:

(A) 0.5

(B) 1.25

(C) 0.75

(D) 1.58 **(Correct Answer)**

Note: Maximum entropy for N classes is $\log_2(N)$. $\log_2(3) \approx 1.58$.
Topic: Lecture 1 & 2 - Entropy

17. In decision tree induction, more split information of an attribute means more gain ratio.

(A) True

(B) False **(Correct Answer)**

Topic: Lecture 3 - Decision Tree Splitting metrics

18. In ID3 algorithm, if for some attribute entropy is maximum then it would be selected as the root node.

(A) True

(B) False **(Correct Answer)**

Note: ID3 selects the attribute with the maximum Information Gain, not maximum entropy.
Topic: Lecture 2 - ID3 Algorithm

19. An attribute may be present in more than one branch of a decision tree.

(A) True **(Correct Answer)**

(B) False

Topic: Lecture 1 - Decision Tree Learning

20. If all samples of a dataset belong to only one class, ID3 would produce a tree with _____ nodes.

(A) 2

(B) 0

(C) 1 **(Correct Answer)**

(D) None of the given options

Note: If the set is pure (entropy = 0), it immediately becomes a single leaf node.
Topic: Lecture 1 & 2 - Decision Tree Leaves

Part II: Subjective Questions

Question 4 (Decision Trees, Entropy, and ID3)

(a) What is the information content, as measured by the Entropy, of a fair coin? How would it change if the coin has been rigged to come up heads 3/4 of the time? [5 Marks]

Student's Solution provided in the scan:

- Entropy of a fair coin would be 1. Because there is a 1/2 chance of both heads and tails.
  $E(S) = -\left(\frac{1}{2}\right)\log_2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)\log_2\left(\frac{1}{2}\right) = 0.5 + 0.5 = 1$
- If the coin is rigged, it means heads has 3/4 chance while tails has 1/4 chance.
  $E(S) = -\left(\frac{3}{4}\right)\log_2\left(\frac{3}{4}\right) - \left(\frac{1}{4}\right)\log_2\left(\frac{1}{4}\right)$
  $E(S) \approx 0.311 + 0.5 = 0.811$
- *Conclusion:* Information content is the measure of uncertainty. A rigged coin has less uncertainty, hence lower entropy.

(b) What is the inductive bias of ID3 algorithm? [2 Marks]

Student's Solution provided in the scan:

- The inductive bias of the ID3 algorithm is that it prefers shorter trees over longer trees. Furthermore, it has a bias towards attributes with many values (like unique record identifiers) because they yield high Information Gain, which is sometimes not feasible (it creates wide, shallow trees that overfit).

(c) We are given a task to build a decision tree classifier for the following representative sample of students data (which has been condensed in two tables for the two classes of students: graduate and undergraduate). Calculate information gain for the attribute 'GPA'. [8 Marks]

Provided Data Table Snapshot (Graduate Students Only):

Gender	Major	Age Range	GPA	Nationality	Count
M	Science	21...25	good	Pakistani	16
F	Science	21...25	excellent	Pakistani	22
M	Engineering	26...30	excellent	Foreigner	18
F	Science	26...30	very good	Foreigner	25
M	Engineering	26...30	excellent	Pakistani	10
F	Engineering	21...25	good	Pakistani	07

Note: The question on the exam required you to calculate the Entropy of the dataset and then subtract the weighted Entropy of the GPA attribute to find the Information Gain.
Topic reference: Lecture 2 - "Calculate entropy for the PlayTennis dataset" & "Information gain"

(d) If unique record identifiers are taken as regular attributes while inducing a decision tree from data, they tend to become the root node due to high 'Gain'. What could be an alternative to using 'Gain' as a metric for decision tree induction? [5 Marks]

Student's Solution provided in the scan:

- The "Gain" method can't be taken in the case of unique record identifiers (like an ID number) taken as regular attributes because their gain would be too high, resulting in them being selected as the root. To solve this problem, we use **Gain Ratio** as an alternative metric for decision trees. Gain Ratio penalizes attributes with a large number of distinct values by taking the "split information" into account, so unique identifiers won't be improperly selected.